List of Publications

The contents of this thesis have been published in the following articles:

- Section 2.5: Adapted from [14], where we solved a problem of Brian and Clontz. I presented the main result at the 2023 Summer Meeting of the Canadian Mathematical Society in Ottawa.
- Chapter 3: Primarily based on my contributions to [9], specifically regarding the separation of learnability classes with computational structures, and the connections to other fields. The contents were presented at the Fields Institute 2026 Set Theory Seminar on January 30th by myself, and continued on February 6th by one of the coauthors. The background was also heavily influenced by a series of four talks we gave at the same seminar in 2024 about Rosenthal compacta.
- Section 4.2: Focuses solely on my contributions to two main algorithms, their analysis and development as detailed in [20]. The implementation is published in [19].
- Section 4.3: Based on both the joint work and the sections and main results I contributed to [10] and [4]. I presented the first results at the 2022 Summer Meeting of the Canadian Mathematical Society in St John's. The material presented in this section is focused on my own generalizations, not found in the currently published body of research.
- Section 4.4: Based on [12], specifically on my contributions to several proofs, conjectures and practical implementations.
- I presented the original proof exhibited in Section 2.2 at the Fields Institute 2025 Set Theory Seminar on March 28th.

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Chapter 1

Introduction

This is the introduction chapter of the thesis.

Chapter 2

Descriptive Combinatorics

This is the descriptive combinatorics chapter of the thesis.

2.1 Background

This is the place to explain how to get from Cantor's classical results on Polish spaces, to the G_0 dichotomy and its applications, to more advanced topics inspired by these dichotomies.

2.2 A concise proof of the L_0 dichotomy

[Removed for pending review]

2.3 A Ramsey-type theorem for E_0 trees

I present a Ramsey-type theorem for E_0 trees, which is a key combinatorial ingredient in ongoing work.

2.4 Inexistence of a basis for digraphs of dichromatic number 3

I answer the open question of whether a basis exists for the class of digraphs with dichromatic number at least 3.

2.5 Uncountable sets and an infinite linear order game

This section addresses a question by Will Brian and Steven Clontz regarding an infinite game of perfect information with two players, called *Baker's game*.

Games have many useful applications in fields like topology and set theory. Leaving many details out of this famous example, the Banach–Mazur game (see [15]) is played by two players who alternate turns picking subsets of ω^ω, forming a descending chain $W_0 \supseteq W_1 \supseteq \cdots$. A previously agreed upon

set $S \subseteq \omega^\omega$ constitutes the winning condition: Player I wins if there is an element $x \in S$ such that $x \in W_n$ for all n; otherwise, Player II wins. A well-known result is that Player II has a winning strategy if and only if S is of the first category. In particular, Player II wins whenever S is countable.

The game I study in this section, Baker's game, is similar: two players take turns selecting real numbers; Player I tries to ensure that some element in the fixed set S remains in between all of these real numbers. In [3], the authors show that Player II has a winning strategy if and only if the set S is countable. It makes sense to play this game in more general linear orders, and so they ask if it is possible for Player II to have a winning strategy on an uncountable set S instead, or if this property somehow only holds in the reals, thus providing a characterization of the reals in these terms.

In collaboration with the other author of [14], we showed the existence of a dense linear order of arbitrary infinite cardinality on which Player II has a winning strategy on **every** set S, including uncountable ones. This construction demonstrates that the outcome of Baker's game is highly sensitive to the order type of the underlying space and not merely a matter of cardinality. Concretely, the Brian–Clontz characterization does not extend to arbitrary dense linear orders.

2.5.1 The Cantor Game

In [1], Matthew Baker introduced the following game on the real numbers. Fix a subset $S \subseteq \mathbb{R}$. Player I starts by picking a real number a_0. Then, Player II picks a real number b_0 such that $a_0 < b_0$. In the n-th turn, Player I picks a real number a_n such that $a_{n-1} < a_n < b_{n-1}$ and then Player II picks a real number b_n such that $a_n < b_n < b_{n-1}$. After ω-many turns they form two sequences of real numbers $\{a_n\}$ and $\{b_n\}$ such that $a_0 < a_1 < \cdots < a_n < \cdots < b_n < \cdots < b_1 < b_0$. Player I wins if and only if there exists $x \in S$ such that $a_n < x < b_n$ for all $n < \omega$. Otherwise, Player II wins. It should be noted that the $[a_n, b_n]$ form a nested set of intervals, and thus there always exists real number x with the property that for all $n < \omega$ $a_n < x < b_n$. Proposition 2.4 generalizes the following observation when playing the game in an arbitrary dense linear order.

Proposition 2.1 ([1])**.** *If S is countable, then Player II has a winning strategy in the Cantor Game.*

The so-called *Cantor Game* was introduced as a game-theoretic way of proving that $\mathbb{R}$ is uncountable, which follows by the previous observation. Baker also proved that if S contains a perfect set, then Player I has a winning strategy. Later in [13], M.D. Ladue (a student of Baker) proved that Player I has a winning strategy if and only if the set S contains a perfect set. The question of whether the converse of Proposition 2.1 is true remained open until 2022. In [3], Brian and Clontz used elementary submodels to prove the converse statement of proposition 2.1 above.

Question 2.1 ([3])**.** Is there a linear order $(X, <)$ and an uncountable $S \subseteq X$ on which Player II has a winning strategy?

2.5.2 Scattered and well-ordered sets

Given an ordinal α, denote by α^* the converse of α, i.e., the conversely well-ordered set $(\alpha, >)$. Recall that a linearly ordered set is *scattered* if it does not contain a copy of the rationals. A linearly ordered set is said to be *σ-scattered* if it is a countable union of scattered sets. Given an order type γ, say that a linearly ordered set is *γ-free* if it does not contain a copy of γ. The following result can be found in [17]. I include the proof for the reader's convenience.

Theorem 2.2 ([16])**.** *Let X be a linearly ordered set. The following are equivalent.*

(i) X is a countable union of well-ordered sets.

(ii) X is σ-scattered and ω_1^*-free.

Proof. $(i) \Rightarrow (ii)$ is clear, so I focus on $(ii) \Rightarrow (i)$. It suffices to assume that X is scattered (and ω_1^*-free). Define the following relation on X: $x \sim y$ if and only if the closed interval $[x, y]$ is a countable union of well-ordered sets. Routine arguments show that $\sim$ is an equivalence relation. Also, every equivalence class $\mathcal{C}$ is convex, i.e., if $x, y \in \mathcal{C}$ and $x < z < y$, then $z \in \mathcal{C}$. Let $[x]$ be the equivalence class of x. Since every equivalence class is convex, the order $[x] < [y]$ if and only if $x < y$ is well-defined. Hence, the quotient order $X/\sim$ is isomorphic to a subspace of X. It follows that $X/\sim$ is scattered by assumption.

Claim 2.2.1. $[x]_\geq := \{y \in [x] : y \geq x\}$ *is a countable union of well-ordered sets for all x.*

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a cofinal subset of $[x]_\geq$ with $y_0 = x$. So,

$$[x]_\geq = \bigcup_{\alpha<\kappa} [y_\alpha, y_{\alpha+1}].$$

Since $[y_\alpha, y_{\alpha+1}]$ is a countable union of well-ordered sets, then write $[y_\alpha, y_{\alpha+1}] = \bigcup_{n<\omega} A_n^\alpha$, where A_n^α is ω^*-free. Let $A_n := \bigcup_{\alpha<\kappa} A_n^\alpha$ and notice that A_n is ω^*-free since κ is ω^*-free. Then, $[x]_\geq = \bigcup_{n<\omega} A_n$, so $[x]_\geq$ is a countable union of well-ordered sets. ⊣

Claim 2.2.2. $[x]_\leq := \{y \in [x] : y \leq x\}$ *is a countable union of well-ordered sets for all x.*

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a coinitial subset of $[x]_\leq$ with $y_0 = x$. Since X is ω_1^*-free, then κ is countable, so

$$[x]_\leq = \bigcup_{n<\omega} [y_{n+1}, y_n].$$

Since $[y_{n+1}, y_n]$ is a countable union of well-ordered sets for all $n < \omega$, then $[x]_\leq$ is a countable union of well-ordered sets. ⊣

The previous two claims imply that $[x]$ is a countable union of well-ordered sets for all $x \in X$. Now I show that actually $X = [x]$ for some x.

Claim 2.2.3. $(X/\sim, <)$ *is dense, i.e., if $[x] < [y]$ then there is $z \in Z$ such that $[x] < [z] < [y]$.*

Proof of Claim. Let $[x] < [y]$ and suppose there is no z such that $[x] < [z] < [y]$. Then, $[x, y] \subseteq [x]_\geq \cup [y]_\leq$ which implies that $[x, y]$ is a countable union of well-ordered sets by the previous two claims. Therefore, $[x] = [y]$. ⊣

This last claim shows that X must have a single equivalence class, since a nontrivial scattered dense order contains a copy of $\mathbb{Q}$. □

If S is a countable union of well-ordered subsets of an ω_1-free set X, then S must be countable. Also, it is clear that a subset S of $\mathbb{R}$ is countable if and only if S is a countable union of well-ordered sets. This idea, in combination with the preceding theorem, is generalized by the following.

Corollary 2.3. *If X is an ω_1-free and ω_1^*-free linearly ordered set, then the following are equivalent for any $S \subseteq X$.*

1. *S is countable.*
2. *S is a countable union of well-ordered sets.*
3. *S is σ-scattered.*

2.5.3 Baker's game on linear orders

I define a variation of Baker's game in an arbitrary dense linear order $(X, <)$. Given $S \subseteq X$, denote by $BG(X, S)$ the game where two players take turns selecting elements a_n, b_n of X as in the Cantor Game for the reals. Player I wins if after ω-many turns there is an $x \in S$ such that for all $n < \omega$, $a_n < x < b_n$, and Player II wins otherwise. The density condition ensures the game never reaches a stalemate. Thus, for the rest of the section, X is a dense linearly ordered set with a fixed subset S (the winning conditions). Observe that every countable set S satisfies the subsequent proposition.

Proposition 2.4. *If S is a countable union of well-ordered sets, then Player II has a winning strategy for $BG(X, S)$.*

Proof. Let $S = \bigcup_{n<\omega} S_n$ where each S_n is ω^*-free (that is, well-founded). In the n-th turn, Player II plays $b_n = \min(S_n \cap (a_n, b_{n-1}))$ if $S_n \cap (a_n, b_{n-1}) \neq \emptyset$ and any legal move otherwise (by density, there must always be one). Now, suppose that for all $n < \omega$, $a_n < x < b_n$. This implies that $x \notin S_n$ for all $n < \omega$, and hence, $x \notin S$. In other words, this is a winning strategy for Player II. □

In general, S being a countable union of well-ordered sets does not imply that S is countable (unless it contains no copies of ω_1). The previous proposition implies that if S is an (uncountable) well-ordered set, then Player II has a winning strategy for $BG(X, S)$. I now show that if S is conversely well-ordered, then Player II has a winning strategy for $BG(X, S)$. Notice that an uncountable conversely well-ordered set S has the property that S cannot be written as a countable union of well-ordered sets, so this result shows that, in general, the converse of Proposition 2.4 is false.

Theorem 2.5. *Suppose that γ is an ordinal and $\{S_\alpha : \alpha < \gamma\}$ is a collection of subsets of X on which Player II has a winning strategy. Furthermore, assume that $S_\beta < S_\alpha$ for all $\alpha < \beta < \gamma$ (i.e. $x < y$ for all $x \in S_\beta$ and $y \in S_\alpha$). Then, Player II has a winning strategy for $BG\left(X, \bigcup_{\alpha<\gamma} S_\alpha\right)$.*

Proof. Player I starts by playing $a_0 \in X$. If there is $b_0 > a_0$ such that $b_0 \leq y$ for all $y \in \bigcup_{\alpha<\gamma} S_\alpha$, then Player II wins by picking b_0. Otherwise, Player II picks any legal move in the first turn. In this case, $a_1 > a_0$, so let $\alpha_1 < \gamma$ be the smallest such that $y_1 < a_1$ for some $y_1 \in S_{\alpha_1}$. If there is a legal move $b_1 > a_1$ such that $b_1 \leq y$ for all $y \in \bigcup_{\alpha<\alpha_1} S_\alpha$, then Player II picks such b_1 and then uses the winning strategy for S_{α_1}. Otherwise, Player II plays any legal move b_1 and then, since $a_2 > a_1$, one can pick the smallest $\alpha_2 < \alpha_1$ such that $y_2 < a_2$ for some $y_2 \in S_{\alpha_2}$.

By repeating this process, since γ is well-ordered, one cannot have a decreasing sequence $\cdots < \alpha_2 < \alpha_1 < \gamma$. Hence, at some turn N, there is a legal move $b_N > a_N$ for Player II such that $b_N \leq y$ for all $y \in \bigcup_{\alpha<\alpha_N} S_\alpha$. After that, Player II continues playing with the winning strategy for $BG(X, S_{\alpha_N})$.

This describes a winning strategy for Player II. Indeed, given any $x \in X$ such that $a_n < x < b_n$ for all $n < \omega$, the following holds: Given that $y_N < a_N < x$ and the hypothesis that $S_\beta < S_\alpha$ whenever $\alpha < \beta$, $x \notin S_\alpha$ for all $\alpha > \alpha_N$; also $x \notin S_\alpha$ for all $\alpha < \alpha_N$ by the way b_N was played. Finally, $x \notin S_{\alpha_N}$, since Player II used a winning strategy for $BG(X, S_{\alpha_N})$. □

Corollary 2.6. *If S is conversely well-ordered, then Player II has a winning strategy on $BG(X, S)$.*

Corollary 2.7. *For any infinite cardinal κ, there is a dense linear order $(X, <)$ of size κ on which Player II has a winning strategy on every subset of X.*

Proof. Take $X = \kappa^* \times \mathbb{Q}$ ordered lexicographically. Concretely, given $x = (\alpha, p)$ and $y = (\beta, q)$ elements of X, $x \leq y$ if either $\beta < \alpha$ or $\alpha = \beta$ and $p <_{\mathbb{Q}} q$. Since the rationals are countable and κ is infinite, X has the desired size. The fact that X is dense follows from the fact that $\mathbb{Q}$ is dense and unbounded.

Now let $S \subseteq X$ and define, for each $\alpha < \kappa$, $S_\alpha = \{\alpha\} \times \mathbb{Q}$. As each S_α is countable, Player II has a winning strategy for $BG(X, S_\alpha)$, by virtue of Proposition 2.4. Since $S = \bigcup_{\alpha<\kappa} S_\alpha$ and the S_α are ordered the way that Theorem 2.5 requires, the result follows. □

2.5.4 Concluding remarks and open questions

When does the converse of Proposition 2.4 hold? By the previous results such orderings must be ω_1-free and ω_1^*-free. Separable and even ccc linear orderings have this property. The separable case (the reals) was established in [3]. Then, the next step is to study dense linear orders that are ω_1-free, ω_1^*-free but not separable. A classical example of such ordering is the Aronszajn ordering.

Definition 2.8. An *Aronszajn line (A-line)* is an uncountable linearly ordered set that is ω_1-free, ω_1^*-free and contains no uncountable subset of the reals.

Following [18], define a proper decomposition for A-lines. Let A be a dense Aronszajn line. Let $D_0 := \emptyset$. Given a countable $D_\alpha \subseteq A$, consider the equivalence relation on $A \backslash D_\alpha$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since D_α is countable and dense in $D_{\alpha+1} := D_\alpha \cup D'_\alpha$, then $D_{\alpha+1}$ must be countable because A does not contain uncountable separable subsets. Given D_α for $\alpha < \gamma$ and γ limit, let $D_\gamma = \bigcup_{\alpha<\gamma} D_\alpha$. Since A does not contain copies of ω_1 and ω_1^*, $A = \bigcup_{\alpha<\omega_1} D_\alpha$. Also, $(T, \supseteq)$ is an Aronszajn tree where $T = \bigcup_{\alpha<\omega_1} T_\alpha$.

Question 2.2. Is there a dense Aronszajn line A where Player II has a winning strategy in $BG(A, S)$ for all $S \subseteq A$?

Another class of lines where a similar decomposition can be performed is the class of nowhere separable Souslin continua.

Definition 2.9. A *Souslin continuum* is a non-separable dense, complete linearly ordered set that has the countable chain condition (ccc), i.e., every family of pairwise disjoint open intervals is countable. A Souslin continuum is *nowhere separable* if no non-empty open interval is separable.

Define a *proper decomposition* for the nowhere separable Souslin continuum L as follows. Let $D_0 := D \neq \emptyset$ be any countable set. Given a countable $D_\alpha \subseteq L$ consider the equivalence relation

on $L\backslash\overline{D_\alpha}$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Again, each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since L has the ccc, then T_α is countable. Let $D_{\alpha+1} = D_\alpha \cup D'_\alpha$. For $\gamma < \omega_1$ limit let $D_\gamma = \bigcup_{\alpha<\gamma} D_\alpha$.

Lemma 2.10. *If $\{D_\alpha : \alpha < \omega_1\}$ is a proper decomposition of a nowhere separable Souslin continuum L, then*

$$L = \bigcup_{\alpha<\omega_1} \overline{D_\alpha}.$$

Proof. Suppose that $x \in L$ and $x \notin \bigcup_{\alpha<\omega_1} \overline{D_\alpha}$. For each $\xi < \omega_1$ let $I^0_\xi := [x]_\xi$ (the $\sim_\xi$ class of x). Note that $I^0_\eta \subseteq I^0_\xi$ for all $\xi < \eta$.

Claim 2.10.1. *For all $\alpha < \omega_1$, $|T_\alpha| > 1$.*

Proof of Claim. Let $x_0 \in D_\alpha$. Since L is nowhere separable, $\overline{D_\alpha}$ is nowhere-dense so pick $y_1 \in (L\backslash\overline{D_\alpha}) \cap (-\infty, x_0)$ and $y_2 \in (L\backslash\overline{D_\alpha}) \cap (x_0, \infty)$. Then, $[y_1]_\alpha, [y_2]_\alpha \in T_\alpha$. ⊣

For each $\xi < \omega_1$ let $I^1_{\xi+1} \in T_{\xi+1}$ such that $I^1_{\xi+1} \cap I^0_{\xi+1} = \emptyset$. Then, $\{I^1_{\xi+1} : \xi < \omega_1\}$ is an uncountable pairwise disjoint family of open sets, contradicting ccc. □

Using a proper decomposition to show the following remains an open question of great interest to the authors of [14].

Conjecture 2.1. Let L be a nowhere separable Souslin continuum. Then Player II has a winning strategy for $BG(L, S)$ if and only if S is countable.

Chapter 3

Deep Computations

This is the deep computations chapter of the thesis.

This work was presented over the course of multiple sessions at the Fields Institute.

3.1 Introduction

3.2 Connections to learning theory

I formalize the connection between deep computations and learning theory. Here I include a series of examples connecting different variants of the NIP. A good explanation or analogy would be great in this section. Expanding on the initial concepts from [9], I explicitly map the relationships between the Non-Independence Property (NIP), VC-dimension, and PAC-learnability within this new context.

3.3 The Bourgain-Fremlin-Talagrand theorems

I synthesize the Bourgain-Fremlin-Talagrand theorems, organizing and refining the material presented during four seminars at the Fields Institute into a cohesive reference for this field. The first parts of the theory were presented on the date**??** and then also on this other day.

3.4 The non-independence property

I analyze the model-theoretic aspects of deep computations, isolating the specific role NIP plays in the combinatorial properties of definable sets.

3.5 Compositional computation structures

This section details my primary contribution to [9]: the construction of concrete examples of computation structures that separate learnability classes. These examples provide the necessary counterpoints to establish the boundaries of the theory.

3.6 Quantum logics

Work in progress...

Chapter 4

Group Actions on Graphs

This is the group actions on graphs chapter of the thesis.

4.1 Introduction

4.2 Border tracing in arbitrary adjacency graphs

While border tracing is standard for finite planar graphs, I generalize these algorithms to arbitrary adjacency graphs.

My Contribution: I developed and analyzed two generalized algorithms for this setting.

4.3 Recursive constructions

4.3.1 Balanceable graphs

Ramsey's classical theorem for finite graphs states that, for a fixed graph G, there is a large enough complete graph K_n such that no matter how the edges of K_n are coloured (in say two colours), one can find a monochromatic copy of G inside of K_n. Suppose one wishes to find a copy of G inside the coloured graph K_n, but instead of monochromatic, with half of the edges (possibly rounding if G has an odd number of edges) in one colour, and the other half in the remaining colour. Clearly, one needs to impose additional conditions on the colourings of K_n, namely, that each colour class is "well represented" (since otherwise one could colour K_n in a single colour, making it impossible to find a non-trivial graph with the desired property) and one needs additional conditions on the structure of G. I do not go into the details of what those properties are, beyond what is needed to state and expand the results from my own research, as well as tie these concepts to the other ideas in this section. The interested reader will find a thorough discussion and complete characterization of these graphs in [6, Section 2].

Definition 4.1. Suppose that a finite graph G has an edge-colouring $c : E(G) \to 2$ in two colours. This coloured graph is called *balanced* if the number of edges of each colour differ by at most one; that is, if G has an even number of edges, there are exactly as many edges of each colour.

A finite graph G is called *balanceable* if in any 2-colouring of the complete graph K_n, with sufficiently many edges of each colour and n sufficiently large, one can find a balanced copy of G.

The *balancing number* $\mathrm{bal}(n, G)$ of a balanceable graph G is the maximum number k of edges for which there is a 2-coloring of K_n with exactly k edges in one colour without balanced copies of G.

In other words, any 2-coloring of K_n with more than $\mathrm{bal}(n, G)$ edges in each color contains a balanced copy of G.

Theorem 4.2 ([6, Corollary 2.5])**.** *A finite graph G is balanceable if and only if:*

1. *there are disjoint $X, Y \subseteq V(G)$ such that $V(G) = X \cup Y$ and there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges between the vertices of X and Y; and*
2. *there is $W \subseteq V(G)$ such that there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges between vertices of W.*

There are more subtle aspects to distinguish in a graph other than whether it is balanceable or not. Interestingly, the balancing number may or may not depend on n, as the following examples demonstrate.

Example 4.3.

1. The path on three edges, P_4, has balancing number 0.

   Suppose that the edges of K_n are coloured in red and blue with at least some edge of each colour. In particular, a red edge and a blue edge must intersect, forming a path of two edges of different colour. Taking any third edge that completes the copy of P_4, successfully builds a balanced copy regardless of the colour of the third edge.

2. The 4-cycle, C_4, has balancing number 1.

   Indeed, it is obvious that the balancing number cannot be less than 1, as C_4 has four edges, and hence a balanced copy needs two of each colour.

   As for the upper bound, let $n \geq 4$ and fix an arbitrary edge colouring $c : E(K_n) \to \{\mathrm{red}, \mathrm{blue}\}$ with at least two edges of each colour.

   First, suppose there is a vertex $x \in V(K_n)$ all of whose neighbours are joined with edges of the same colour (say red). Then, if there are two blue edges in K_n that intersect, they form a blue path of vertices p_0, p_1, p_2, neither of which can be x by assumption. The edges $(x, p_0), (p_0, p_1), (p_1, p_2), (p_2, x)$ are red, blue, blue, red, respectively, and thus form a balanced copy of C_4. If no two blue edges intersect, then the edges joining the endpoints of any two blue edges (of which there must be at least two) must be all red, and therefore one can form a balanced copy of C_4.

   Now, consider the case when every vertex has red and blue incident edges. Continue by selecting a red edge xy and partitioning $V(K_n) \setminus \{x, y\}$ into four classses depending on the value of the pair $(c(wx), c(wy))$. There are two colours, hence four possible values for this pair; let us abbreviate the classes as RR for those vertices w with pair $(\mathrm{red}, \mathrm{red})$ and similarly, RB, BR and BB. By assumption, some blue edge is incident in x and thus $BR \cup BB \neq \emptyset$; similarly, $RB \cup BB \neq \emptyset$. The same reason says that $V(K_n) \setminus \{x, y\}$ cannot be contained in a

single class from RB, BR or RR. If $V(K_n) \setminus \{x, y\} = BB$, then any red edge different from xy must be disjoint from xy and forms a balanced copy of C_4 by connecting said edge to xy (via blue edges). Therefore, there must be at least two nonempty classes. If $w \in RR$ and $w' \in BB$, or if $w \in RB$ and $w' \in BR$, then x, w, y, w' form a balanced 4-cycle.

Let's assume that $w \notin BB$ **??**.

3. The graph consisting of four disjoint edges, $4K_2$, has balancing number $n - 1$. Concretely, when $n \geq 10$, $\mathrm{bal}(n, 4K_2) = n - 1$. This is [8, Theorem 2.3].

4. When $n \geq 5$, $\mathrm{bal}(n, K_{1,4}) = n - 1$. This is an immediate consequence of [6, Theorem 3.2].

5. The complete graph K_5 is not balanceable.

   Consider a colouring of K_{2n} where we split the vertex set into two disjoint sets A and B of size n. We colour every edge connecting vertices in A red, and edges connecting vertices in B blue; edges connecting A with B are also red. Suppose a balanced copy C of K_5 sits in K_{2n} with vertex set $V(C) \subseteq V(K_{2n})$; then five of its ten edges must be blue. The vertex set $V(C) \cap B$ induces a complete graph (a clique in K_{2n}) such that an edge $xy \in E(C)$ is blue if and only if $\{x, y\} \subseteq B$. Since no complete graph has five edges, this is a contradiction.

   This example illustrates the characterization provided by Theorem 4.2. The preceding argument shows that that K_5 fails the second condition of the theorem; it is straightforward (albeit unnecessary) to show that it also fails the first.

Let me elucidate the case when the value of $\mathrm{bal}(n, G)$ is independent of the parameter n. So, suppose that G is a graph on k edges (for simplicity, let's say k is always even), such that $\mathrm{bal}(n, G) = C(G)$. In such a case, I say that G has *constant balancing number*, such as in Example 4.3(1). The graph $\frac{k}{2}K_2$ is a set of $k/2$ disjoint edges (a *matching*), and $K_{1,k/2}$ is a *star graph*, that is, $k/2$ edges incident on a single vertex. Now, fix the complete graph K_n. One can always pick an arbitrary vertex v_0 and colour all edges incident in v_0 red and every other edge of K_n blue, as done in Example 4.3(2). Provided one makes n large enough, one can guarantee that K_n has more than $C(G)$ red edges. By definition, there is a balanced copy of G inside K_n. In particular, all of its red edges (which account for exactly $k/2$ edges of G) intersect at some vertex. This vertex in G can be then chosen as the image of the center of the star graph $K_{1,k/2}$ under an injective graph homomorphism $K_{1,k/2} \to G$.

Consider a different colouring of K_n. This time, pick a maximal matching in K_n and colour those edges red, and everything else blue. The size of the matching increases with n, and so once again, for large enough n, we can guarantee that G satisfies Definition 4.1. Consequently, (at least) half of the edges of G must be pairwise disjoint. In other words, one can construct an injective homomorphism $\frac{k}{2}K_2 \to G$.

As it turns out, these two aforementioned properties fully characterize graphs of constant balancing number. In [4], we proved this result in parallel to the same characterization published in [7, Section 3.2], where the value of $C(G)$ found involves Ramsey numbers, which grow exponentially. In our paper, the characterization offers a deeper understanding of the structure of the graphs with constant balancing number, and furthermore shows that $\mathrm{bal}(n, G)$ is quadratic in k (and not just bounded by a function exponential in k). Moreover, we extend the characterization to graphs with an odd number of edges.

Theorem 4.4. *Let $k > 0$ be an even integer and G a graph with k edges and no isolated vertices. Then, the following are equivalent.*

1. *G has constant balancing number.*
2. *There are injective graph homomorphisms $\frac{k}{2}K_2 \to G$ and $K_{1,k/2} \to G$.*
3. $$\mathrm{bal}(n, G) \leq \frac{(|V(G)| - 3)(|V(G)| + k)}{2}.$$

Moreover, if G has either k or $k+1$ edges, then

$$\frac{1}{18}k^2 - \frac{11}{18}k + \frac{14}{9} \leq \mathrm{bal}(n, G) \leq \frac{15}{8}k^2 - \frac{7}{4}k - 1.$$

Implication (1) $\Longrightarrow$ (2) is [7, Theorem 3.6] proved using Ramsey numbers and also [4, Theorem 1.1] proved by an involved structural induction. The implication (2) $\Longrightarrow$ (3) is [4, Theorem 3.7] and is the only part that requires that G has no isolated vertices. The moreover part is [4, Theorem 5.1]. The implication (3) $\Longrightarrow$ (1) is immediate.

If one wishes to study an analogue notion for infinite graphs, one has to define what it means for a large complete graph to have "sufficiently many edges of each colour." In [2, Section 4], the authors study a somewhat similar (albeit different) notion of balanceability and suggest an approach to generalizing these ideas to the infinite case. I draw inspiration from this approach and adapt some of the ideas to our theory. Namely, that "sufficiently many edges of each colour" corresponds to non-meagre colour classes in a Polish space. The basic notions regarding graphs defined on Polish spaces is covered in Section 2.1.

I prove an analog of Theorem 4.4 that says that if the colours are well represented in a complete graph on a perfect Polish space, then one can find a star graph: a fixed centre connected to a Cantor set through edges in one colour, joined with perfectly-many pairwise disjoint edges in the other colour; I believe this to be the right interpretation of the uncountable version of a graph with constant balancing number.

Let X be a perfect Polish space.

Lemma 4.5. *Given non-meagre disjoint sets $R, B \subseteq [X]^2$ with the Baire property, there are disjoint open sets $U_R, U_B \subseteq X$ such that R and B are comeagre in $[U_R]^2$ and $[U_B]^2$, respectively.*

Proof. □

Theorem 4.6. *Suppose $\chi : [X]^2 \to 2$ is a Baire measurable function with non-meager fibres (the colour classes). Then, there is a colour $i < 2$, a vertex $c \in X$, and a Cantor set $C \subseteq X$ such that for all $x \in C$, $\chi(\{c, x\}) = i$, and such that $[C]^2$ contains an uncountable matching of colour $1 - i$.*

Proof. □

A large family of graphs with strong balanceability properties is defined and explored in Section 4.3.2.

4.3.2 The group of feasible edge-replacements

Consider this combinatorial puzzle: a graph is given; one may remove an edge and add it wherever there is no edge (trivially, one can add it back to its original position), but the new graph must be isomorphic to the previous graph. Such an action is called a *feasible edge-replacement*, which is formalized after these examples. Can one attain every possible configuration of the graph by repeateadly changing edges like this? It is clear that the answer depends on the structure of the particular starting graph. For instance, a cycle of four edges has no valid non-trivial movements since, having removed an edge, the only possibility that preserves the cycle is putting the edge back from where it was taken. Graphically, Figure 4.1 depicts two isomorphic graphs that cannot be transformed into each other by feasible edge-replacements.

Figure 4.1: Two 4-cycles on the same vertex set.

The two paths in Figure 4.2 are isomorphic. Is it possible to transform one into the other by a sequence of feasible edge-replacements?

Figure 4.2: Two paths of length 3 on the same vertex set.

For a set X, denote by S_X the set of all bijections (also called *permutations*) $X \to X$. Endow this set with the topology inherited from the Tychonoff product X^X, with X taken as discrete. Let G be a graph defined on a set V with a labeling $\lambda : V \to X$. When clear from context, I will identify a permutation $\sigma \in S_X$ with the induced mapping $V \to V$ given by $\lambda^{-1} \circ \sigma \circ \lambda$.

Suppose that $\overline{G}$ denotes the graph complement of G (that is, the graph obtained from switching every adjacency and every non-adjacency in G) and let $(e, f) \in E(G) \times E(\overline{G})$. I use the phrase "$\sigma : (e \to f)$ is a *feasible edge-replacement*" to abbreviate the fact that σ is a graph isomorphism between G and $G - e + f$. If σ is an automorphism of G, I write $\sigma : (\emptyset \to \emptyset)$ (I employ the symbol $\emptyset$ to signal that no edge is being moved). Denote by $\mathrm{Fer}(G)$ the group generated by all σ for which there are e and f such that $\sigma : (e \to f)$ is a feasible edge-replacement. These σ are called *canonical generators* of $\mathrm{Fer}(G)$. Canonical generators include automorphisms and permutations that correspond to moving one edge. The group $\mathrm{Fer}(G)$ contains elements that can correspond to a *sequence* of edge movements. Notice also that the automorphism group $\mathrm{Aut}(G)$, seen as permutations, is a subgroup of $\mathrm{Fer}(G)$, which is itself a subgroup of S_X.

It is convenient to represent a sequence of feasible edge-replacements by multiplying their corresponding permutations (which, as standard in Algebra, is written right-to-left) and concatenating the list of edges in the order they should get replaced (which, as standard in English, is written left-

to-right). This requires defining some more symbols. There is however a technical caveat: the labels of the graph's vertices change after each canonical generator gets applied (since the original labels get permuted); but having a list of edge replacements in the graph's original labeling is desirable, for example, if one wishes to execute a solution to a puzzle by purely moving edges without having to keep track of what the corresponding permutations are (not to mention the extra effort needed to figure out what those permutations are in the first place). One way to circumvent this is to define the multiplication abstractly by means of recursion (by "abstractly," I mean that defining the operation does not actually require knowledge of the graph G on which it is defined). The following recursive definition is crucial to carrying out these calculations in an object-oriented programming framework.

Definition 4.7. Suppose that x, y, z, w, a, b, c, d, and possibly adding subscripts, are elements (labels of vertices of some graph G) of some set X. If $\sigma, \rho \in S_X$, I declare the symbol $\sigma : (\emptyset \to \emptyset)$ to be a `fer` *object*. If

$$\begin{aligned} &\sigma : (x_0y_0 \to w_0z_0)(x_1y_1 \to w_1z_1)\cdots(x_ny_n \to w_nz_n) \qquad \text{and} \\ &\rho : (a_0b_0 \to c_0d_0)(a_1b_1 \to c_1d_1)\cdots(a_mb_m \to c_md_m) \end{aligned}$$

are `fer` objects, then the `fer` *product* of the two (important: in the order they are written), is the symbols

$$\begin{aligned} \rho\cdot\sigma : (x_0y_0 \to w_0z_0)(x_1y_1 \to w_1z_1)\cdots \\ \cdots(x_ny_n \to w_nz_n)(\sigma^{-1}(a_0)\sigma^{-1}(b_0) \to \sigma^{-1}(c_0)\sigma^{-1}(d_0))(\sigma^{-1}(a_1)\sigma^{-1}(b_1) \to \sigma^{-1}(c_1)\sigma^{-1}(d_1))\cdots \\ \cdots(\sigma^{-1}(a_m)\sigma^{-1}(b_m) \to \sigma^{-1}(c_m)\sigma^{-1}(d_m)) \end{aligned}$$

which I declare to be a `fer` object.

So that this definition becomes useful not just for a programmer but for studying the mathematical objects in question, it is imperative to observe that the preceding abstract definition is in fact consistent with the operations we are interested in.

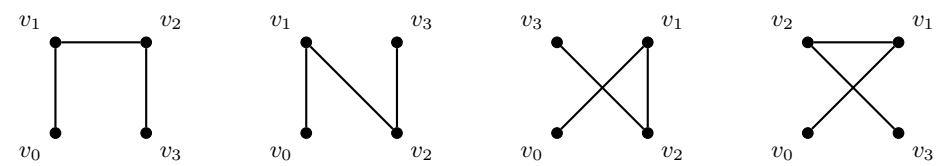
Remark 4.8. Let G be a graph and suppose that $\sigma : (xy \to wz)$ and $\rho : (ab \to cd)$ are feasible edge-replacements. Then the product of these two as `fer` objects is a sequence of feasible edge-replacements, where $(xy \to zw)(\sigma^{-1}(a)\sigma^{-1}(b) \to \sigma^{-1}(c)\sigma^{-1}(d))$ is the corresponding sequence of edge-replacements needed to get from G to the copy of G on V obtained after applying the permutation $\rho\cdot\sigma$ to the labels of its vertices.

For instance, a solution to the puzzle is shown in Figure 4.3, where each caption shows the canonical generator needed to obtain that step of the solution; and the element of $\mathrm{Fer}(G)$ that solves the puzzle (with the same labels as in the figure) is the product of the three generators, resulting in

$$(2\ 3) : (1\ 2 \to 1\ 3)(0\ 1 \to 0\ 2)(2\ 3 \to 1\ 2).$$

It is worth noting that the permutation $(2\ 3)$ is not a canonical generator, as two edges necessarily change position.

For the remainder of the chapter, topological closures are always taken in the containing space X^X. In symbols, if $A \subseteq X^X$, $\overline{A} := \mathrm{cl}(A, X^X)$.



(a) Starting graph (b) $(2\ 3) : (1\ 2 \to 1\ 3)$ (c) $(1\ 3) : (0\ 1 \to 0\ 3)$ (d) $(2\ 3) : (1\ 2 \to 1\ 3)$

Figure 4.3: A sequence of feasible edge-replacements.

Definition 4.9 (Local Amoeba)**.** A graph G is called a *local amoeba* if $\overline{\mathrm{Fer}(G)} = S_X$.

Hence, a local amoeba is precisely a graph that can be transformed into any copy of itself on the same vertex set by a (possibly infinite) sequence of feasible edge-replacements. This graph-theoretic notion and the name *amoeba* first appeared in [6] as an interesting combinatorial object with a property that I briefly describe in Subsection 4.3.1. When X is finite, all subsets of X^X are closed and thus the preceding definition simplifies to $\mathrm{Fer}(G) = S_X$, which is the algebraic notion introduced in [5] as a tool to study these graphs and their properties. The currently published body of research studies the case when X is finite.

If C_4 and P_4 are the examples depicted in Figure 4.1 and Figure 4.2, respectively, then $\mathrm{Fer}(C_4) = \mathrm{Aut}(C_4)$ is the dihedral group D_4 of order $2 \cdot 4$, and $\mathrm{Fer}(P_4) = S_4$ is the symmetric group. Therefore, C_4 is not a local amoeba, while P_4 is.

Complete graphs are all local amoebas for a rather uninteresting reason: their automorphism group is S_X (any permutation preserves the adjecencies trivially, since any two distinct vertices are adjacent). I call these graphs *rigid*.

Definition 4.10. The graph G is *rigid* if $\overline{\mathrm{Fer}(G)} = \mathrm{Aut}(G)$.

The graph G is called *asymmetric* if $\mathrm{Aut}(G)$ is the trivial group consisting of the identity on X.

One of the main concerns of this active field of research is finding and classifying interesting examples of graphs that are local amoebas. For instance, constructing trees with arbitrary large maximum degree that are local amoebas is a curious problem, whose solution was the main precursor of [10], which I present in Section 4.3.3.

In terms of Group Theory, $\mathrm{Fer}(G)$ carries all of the expressive power of finite groups. Precisely, we can show that there is no theorem of the form "for every finite graph G, the group $\mathrm{Fer}(G)$ has property φ," where φ is a non-trivial property about groups.

After Frucht's 1939 breakthrough showing that every finite group is the automorphism group of some finite graph, he extended this result to the more restrictive class of cubic graphs (graphs in which every vertex has degree 3) in 1949 (see [11]). This result, often referred to as Frucht's Theorem, demonstrates that for any finite group G, there exists a cubic graph X such that $\mathrm{Aut}(X) \cong G$.

For a cubic graph G, and more generally for a k-regular graph with $k \geq 2$, G is rigid, that is $\mathrm{Fer}(G) = \mathrm{Aut}(G)$. This is a direct consequence of the fact that applying a non-trivial feasible edge-replacement changes the degree of at least one vertex (by 1). Put succintly:

Proposition 4.11. *For every finite group Γ, there is a graph G such that Γ is isomorphic to* $\mathrm{Fer}(G)$.

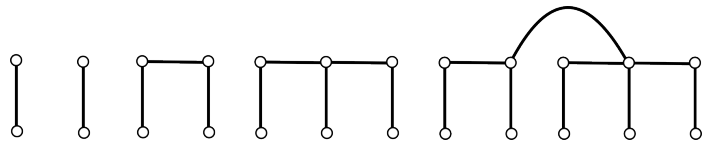


Figure 4.4: T_k for $1 \leq k \leq 5$.

4.3.3 Local amoeba trees with large maximum degree

Intuitively, if one is trying to transform one graph into another by means of feasible edge-replacements, it may come in handy to have isolated vertices that allow new positions for graphs. I previously mentioned that complete graphs are rigid graphs, meaning they have no non-trivial feasible edge-replacements. Adding isolated vertices to a complete graph destroys its rigidity.

Definition 4.12. A finite graph G is called a *global amoeba* if there is a $t \geq 1$ such that $G \cup tK_1$ (that is, the graph obtained as the disjoint union of G and t many isolated vertices) is a local amoeba.

Perhaps surprisingly, adding isolated vertices does not always increase the amount of feasible edge-replacements; in fact, there is no general logical implication between a graph G being a local amoeba and being a global amoeba. One case where such an implication does exist is for trees (see Corollary 4.14).

The main result of [4] is the following classification of global amoebas. Notably, if a value of t exists such that Definition 4.12 holds, then $t = 1$ is the least such value.

Theorem 4.13 ([4, Theorem 4.1])**.** *For any non-empy graph G, the following are equivalent.*

1. *G is a global amoeba.*
2. *$G \cup K_1$ is a local amoeba.*
3. *Every orbit of* $\mathrm{Fer}(G)$ *has a label that corresponds to some vertex of degree 1.*

A well-known fact about finite trees is that they have at least two vertices of degree 1. Joined with the observation that if T is a local amoeba, then $\mathrm{Fer}(T)$ consists of a single orbit (it is in fact, the whole symmetric group), it is clear that condition (3) of Theorem 4.13 is satisfied.

Corollary 4.14. *If T is a finite tree that is a local amoeba, then it is a global amoeba.*

The connectedness of T is indispensable. For example, a matching (a set of pairwise disjoint edges) forms a global amoeba, since every vertex has degree 1, but not a local amoeba (it is, in fact, rigid).

The construction I describe now (see Figure 4.4), loosely inspired by the Fibonacci sequence, illustrates the main results of [10]. Start by defining T_1 and T_2 to each be the tree consisting of a single edge, in other words, $T_1 \simeq T_2 \simeq K_2$. Now, suppose that T_{k-1} and T_{k-2} have already been constructed. In each of these trees, select a vertex of maximum degree. The tree T_k is the result of joining T_{k-1} and T_{k-2} by the edge connecting these two vertices. The choice of the vertex of maximum degree is irrelevant, as one can easily verify: either such vertex is unique, or there are exactly two such vertices that are similar (that is, they can be interchanged via an automorphism), and thus the resulting sequence of graphs is unique up to isomorphisms.

The sequence of graphs $\{T_k : k \in \mathbb{N}\}$ is a particular case of a more general scheme of recursive constructions studied in [10, Section 3].

Theorem 4.15 ([10, Theorem 9])**.** *For every $k \in \mathbb{N}$, T_k is a local amoeba.*

An independent contribution of mine answers the following question which I developed in [10, Section 5]. The proof that a finite graph G is a local amoeba often involves a counting argument: one shows that $\mathrm{Fer}(G)$ has $|X|!$ elements and then, given that $\mathrm{Fer}(G)$ is contained in the symmetric group on X, $\mathrm{Fer}(G)$ must be equal to S_X. There are two disadvantages with this method.

- Given an arbitrary permutation $\sigma \in S_X$ (that is, an arbitrary copy of G on the same vertex set), the proof generally offers no method of finding the sequence of feasible edge-replacements. In other words, this proving method shows the existence of a solution to the puzzle, without giving an algorithm to find it. The algorithm presented in this section resolves this for an infinite family of trees.
- Counting methods don't generalize to infinite graphs. I address this in Section 4.5.

4.4 Amoeba trees

I present the properties of amoeba trees and recent advancements regarding their conjectures.

My Contribution: I independently conceptualized several key properties discussed in this section. Additionally, I authored abstractions and object-oriented frameworks that had a significant impact on the results presented in [12]. Apart from my immediate contributions, I focus on unpublished ideas that helped shaped the research.

4.5 Infinitary amoeba graphs

I synthesize ideas on infinite graphs, focusing on the interaction between automorphism groups, ends, and amenable groups, specifically placing my own results and ideas concerning amoeba graphs within this broader theoretical context.

Chapter 5

Closing Remarks and Open Problems

This is the closing remarks and open problems chapter of the thesis.

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